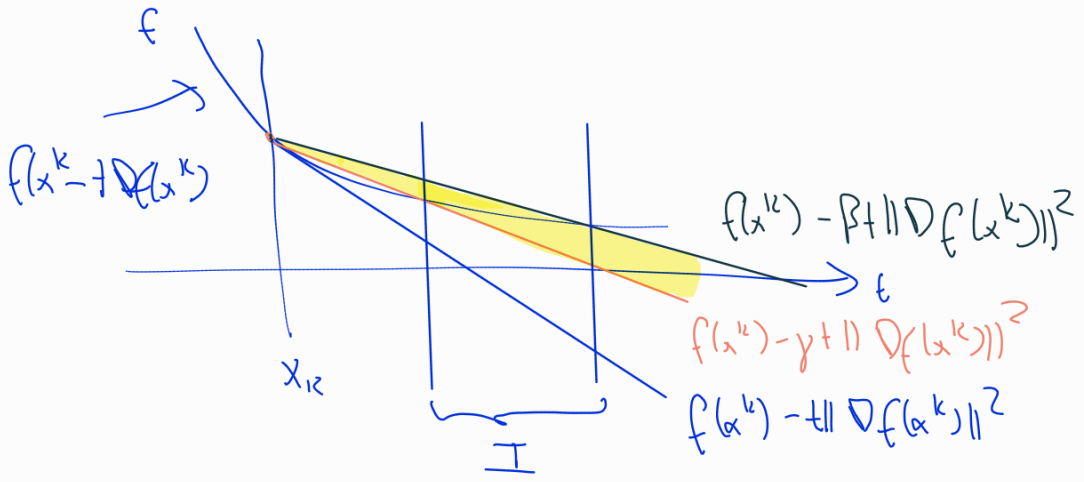
Illustration:



I is the set of possible t according to the Goldstein rule

Interpretation of inequalities:

$f(x^k - t_k \nabla f(x^k)) \leq f(x^k) - \beta t_k \|\nabla f(x^k)\|^2$ $\rightsquigarrow$ appropriate descent

$\geq f(x^k) - \gamma t_k \|\nabla f(x^k)\|^2$ $\rightsquigarrow$ t_k not too small

$\rightsquigarrow$ some kind of convergence

Backtracking step-size rule

04/16/2026

Choose $t_{max} > 0$ and $j \in \mathbb{N}_0$ such that

for $\alpha \in]0,1[$

$$f(x^k - \alpha^j t_{max} \nabla f(x^k)) \leq f(x^k) - \beta \alpha^j t_{max} \|\nabla f(x^k)\|^2$$

and j is minimal. Then: $t_k = \alpha^j t_{max}$

"Algorithm":

for $j = 0, 1, 2, \ldots$

  if $f(x^k - \alpha^j t_{max} \nabla f(x^k)) \leq f(x^k) - \beta \alpha^j t_{max} \|\nabla f(x^k)\|^2$

  then STOP and return j

Stopping criteria

- **Fixed number of iterations**: STOP if $k = k_{max}$ for chosen $k_{max} \in \mathbb{N}$.

↝ Choose k_{max} wisely.

- **Objective function**: Choose $\varepsilon > 0$ tolerance.

STOP if $f(x^k) - f(x^{k+1}) < \varepsilon$

Then: $-\varepsilon < f(x^{k+1}) - f(x^k) \leq -\frac{1}{2L}\vartheta_k(2-\vartheta_k)\|\nabla f(x^k)\|^2$

$$\Rightarrow \|\nabla f(x^k)\|^2 \leq \frac{2\varepsilon L}{\vartheta_k(2-\vartheta_k)} \leq \frac{2\varepsilon L}{c}$$ for some $c > 0$ determined by $(t_k)_k$

It follows that $\|\nabla f(x^k)\|$ is "small",

↝ measure of stationarity

↝ $\min_{m=0,\dots,k-1} \|\nabla f(x^m)\| = O(\frac{1}{k})$, we actually stop at some k

> Recall:
> x^* stationary $\Rightarrow \|\nabla f(x^*)\| = 0$

Related: - **Residual norm**: Choose $\varepsilon > 0$, STOP if

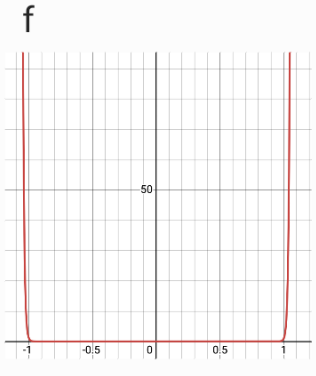
$$\|\nabla f(x^k)\| < \varepsilon$$

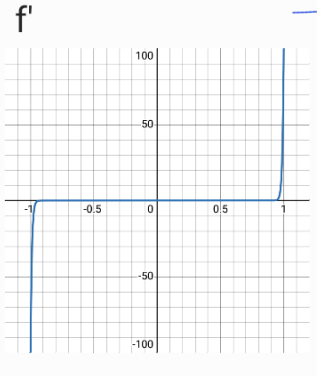
↝ other words, is weaker than the criterion for the objective function

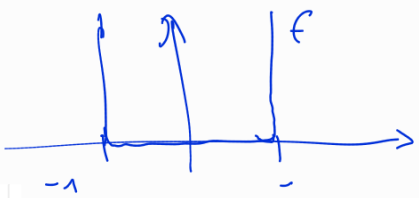
↝ We cannot conclude from $\|\nabla f(x^k)\|$ "small" that $\|x^k - x^*\|$ is "small" if $x^k \xrightarrow{k\to\infty} x^*$ stationary point

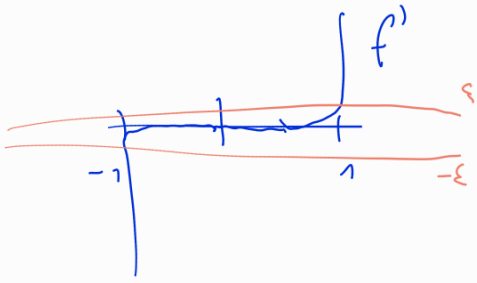
Example: $f(x) = x^{100}$

$f'(x) = 100\,x^{99}$









↝ if ∇f^{-1} exists and is Lipschitz continuous,

then $\|\nabla f(x) - \nabla f(y)\| \geq c\|x-y\| \quad \forall x,y \in \mathbb{R}^n$

and some $c>0$, then: $\|x^k - x^*\| \leq \frac{1}{c}\|\nabla f(x^k)\|$ ← this can be controlled

For instance, if the inverse function theorem is applicable for ∇f, the we obtain this property locally.

Also, if f is strongly convex, then this property is satisfied.

- Change per iteration: Choose $\varepsilon>0$ and STOP if $\|x^{k+1} - x^k\| < \varepsilon$

↝ Recall: $x^{k+1} = x^k - t_k \nabla f(x^k)$ ↝ $\|x^{k+1} - x^k\| = t_k \|\nabla f(x^k)\|$

this could be small

↝ could be misleading if step-size is too small (in practice, slowness is indistinguishable from convergence)

↝ Still: if t_k is constant, we can relate $\|x^0 - x^1\|$ with $\|x^{k+1} - x^k\|$

↝ $\|x^{k+1} - x^k\| \leq \varepsilon \|x^0 - x^1\| \Rightarrow$ STOP ↝ equivalent to residual norm

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Convergence of gradient descent for convex smooth problems

Situation: Let $f: \mathbb{R}^n \to \mathbb{R}$ be convex, bounded from below, but ∇f be L-Lipschitz continuous

Smooth convex optimization problem:

$$\min_{x \in \mathbb{R}^n} f(x)$$

Gradient descent as before: $\liminf_{k\to\infty} t_k > 0$, $\limsup_{k\to\infty} t_k < \frac{2}{L}$.

- If f is coercive, then it has a minimizer $x^* \in \mathbb{R}^n$

$\Rightarrow (x^k)_k$ gradient descent sequence is bounded

$\Rightarrow$ subsequential convergence to a stationary point $\bar{x} \in \mathbb{R}^n$.

Claim: $\bar{x}$ is a minimizer. $f(\bar{x}) + \underbrace{\langle \nabla f(\bar{x}), x - \bar{x}\rangle}_{=0} \leq f(x) \quad \forall x \in \mathbb{R}^n$ (convexity)

$\Rightarrow f(\bar{x}) \leq f(x) \quad \forall x \in \mathbb{R}^n \Rightarrow \bar{x}$ is a minimizer

We have: Subsequential convergence to minimizers

If minimizers are unique, then $x^k \xrightarrow{k\to\infty} x^*$.

Remark: $(x^k)_k$ always converges (but we need other tools for showing this)

Descent property for $t_k = \frac{1}{L}$: $f(x^{k+1}) \leq f(x^k) - \frac{1}{2L}\|\nabla f(x^k)\|^2$

if x^* is a minimizer, also recall $f(x^k) + \langle \nabla f(x^k), x^* - x^k\rangle \leq f(x^*)$

$\leq f(x^*) + \langle \nabla f(x^k), x^k - x^*\rangle - \frac{1}{2L}\|\nabla f(x^k)\|^2$

complete the square

$$= f(x^*) + \frac{L}{2}\|x^k - x^*\|^2 - \frac{L}{2}\|x^k - x^*\|^2 + \frac{L}{2}\frac{2}{L}\langle \nabla f(x^k), x^k - x^*\rangle - \frac{L}{2}\frac{1}{L^2}\|\nabla f(x^k)\|^2$$

$$= f(x^*) + \frac{L}{2}\|x^k - x^*\|^2 - \frac{L}{2}\left[\|x^k - x^*\|^2 - \frac{1}{L}\langle \nabla f(x^k), x^k - x^*\rangle + \frac{1}{L^2}\|\nabla f(x^k)\|^2\right]$$

$$= f(x^*) + \frac{L}{2}\|x^k - x^*\|^2 - \frac{L}{2}\underbrace{\left\|x^k - x^* - \frac{1}{L}\nabla f(x^k)\right\|^2}_{= x^{k+1} - x^*}$$

$$= f(x^*) + \frac{L}{2}\left[\|x^k - x^*\|^2 - \|x^{k+1} - x^*\|^2\right]$$

Sum this up from $m = 0, \dots, k-1$:

$$\sum_{m=0}^{k-1} f(x^{m+1}) - f(x^*) \leq \frac{L}{2}\left[\sum_{m=0}^{k-1} \|x^m - x^*\|^2 - \|x^{m+1} - x^*\|^2\right]$$

$$= \frac{L}{2}\left[\|x^0 - x^*\|^2 - \|x^k - x^*\|^2\right]$$

$$\Rightarrow \left[\sum_{m=0}^{k-1} \underbrace{\left(f(x^{m+1}) - f(x^*)\right)}_{\geq 0} + \underbrace{\frac{L}{2}\|x^k - x^*\|^2}_{\geq 0}\right] \leq \frac{L}{2}\|x^0 - x^*\|^2$$

Further: Since $f(x^{m+1}) \geq f(x^k)$ for each $m = 0, \dots, k-1$

$$\Rightarrow \underbrace{\left[\sum_{m=0}^{k-1} f(x^k) - f(x^*)\right]}_{= k\left[f(x^k) - f(x^*)\right]} \leq \sum_{k=0}^{m-1} f(x^m) - f(x^*) \leq \frac{L}{2}\|x^0 - x^*\|^2$$

In total: $f(x^k) - f(x^*) \leq \frac{L}{2k}\|x^0 - x^*\|^2 = O\left(\frac{1}{k}\right)$

Consequently, if an estimate $\|x^0 - x^*\|^2 \leq M$ is known, then

$f(x^k) - f(x^*) < \varepsilon$ if $\boxed{k > \frac{2M}{L\varepsilon}}$ for $\varepsilon > 0$.

$\Rightarrow$ guaranteed near-optimality w.r.t. objective function.

Remark: This is fine but also shows that gradient descent is slow.

Theorem 2: If f is convex, L-smooth and there is a minimizer $x^* \in \mathbb{R}^n$, then gradient descent with step-size $t_k = \frac{1}{L}$ gives: $f(x^k) - f(x^*) \leq \frac{L}{2k}\|x^0 - x^*\|^2 = O\left(\frac{1}{k}\right)$

Question: What happens if we have additional information?

E.g. Let f be μ-convex for $\mu > 0$.

$$\Rightarrow \quad f(x) + \langle \nabla f(x), y - x \rangle + \frac{\mu}{2}\|x - y\|^2 \leq f(y) \qquad \forall\, x, y \in \mathbb{R}^n$$

↑ additional term due to strong convexity

It follows that:

$$f(x) + \left[\min_{y \in \mathbb{R}^n} \langle \nabla f(x), y - x \rangle + \mu \frac{\|x - y\|^2}{2}\right] \leq f(y)$$

compute this minimum

Differentiation: $0 = \nabla f(x) + \mu(x - \bar{y}) \Rightarrow \bar{y} = x - \frac{1}{\mu}\nabla f(x) \Rightarrow \bar{y}$ minimizer since function is convex

Plug this in:

$$f(x) + \langle \nabla f(x), -\frac{1}{\mu}\nabla f(x) \rangle + \frac{\mu}{2}\left\|\frac{1}{\mu}\nabla f(x)\right\|^2 \leq f(y)$$

$$\Rightarrow \quad f(x) - \frac{1}{\mu}\|\nabla f(x)\|^2 + \frac{1}{2\mu}\|\nabla f(x)\|^2 \leq f(y)$$

$$\Rightarrow \quad f(x) - \frac{1}{2\mu}\|\nabla f(x)\|^2 \leq f(y)$$

Rearrange: $\|\nabla f(x)\|^2 \geq 2\mu(f(x) - f(y))$ $\rightsquigarrow$ leads to better estimates